

Consider a region of the inner leaflet of a cell membrane in which phospholipid components help bind positively charged cytosolic molecules. What can be predicted about the identity and protonation state of the phospholipid components that most likely contribute to the binding?

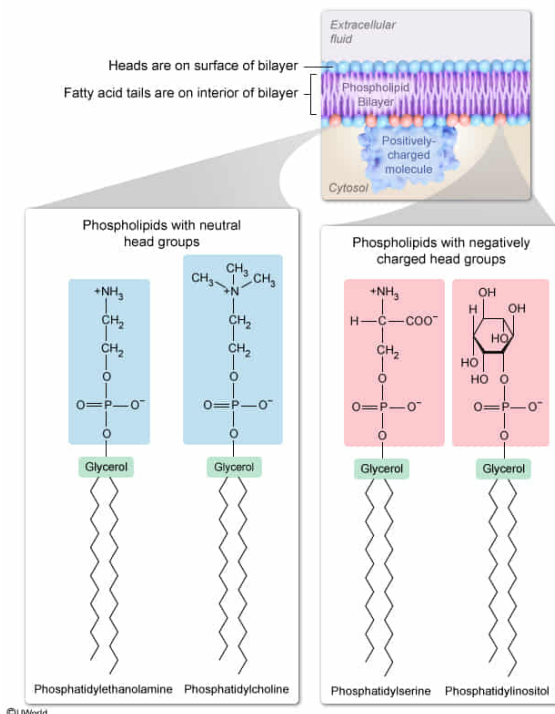
- ✔ ☒ A. Binding must occur between the molecule and phospholipid heads with more deprotonated than protonated functional groups. [69%]
- ☐ B. Binding must occur between the molecule and phospholipid heads in which all functional groups are fully protonated. [10%]
- ☐ C. Binding could occur between the molecule and phospholipid tails or heads that have more deprotonated than protonated functional groups. [14%]
- ☐ D. Binding could occur between the molecule and phospholipid tails or heads that have more protonated than deprotonated functional groups. [5%]

Correct

70% answered correctly

Explanation:

Negatively charged polar head groups promote binding of positively charged molecules to the cell membrane



The membrane enclosing a cell (ie, the plasma membrane) consists of a **phospholipid bilayer** along with various other molecules, as described by the **fluid mosaic model**. The phospholipids comprising the bilayer are **amphipathic**, and this property determines phospholipid orientation in the membrane, with the hydrophilic head groups facing aqueous environments and the hydrophobic tails in the bilayer interior.

The question describes binding of positively charged cytosolic molecules to the inner leaflet of the cell membrane and asks for the predicted identity and protonation state of

the phospholipid component contributing to the binding. Inner leaflet **head groups** are the phospholipid components **facing** the **cytosol**. Protons contribute positive charges to molecules. Therefore, when **more functional groups** in a **phospholipid head** are **deprotonated than protonated**, the head group carries an **overall negative charge** and **attracts** positive molecules, facilitating binding.

(Choice B) If all functional groups in a phospholipid head were fully protonated, the head would carry an overall positive charge that would *repel* rather than attract positively charged molecules. The phosphate group of the phospholipid head is likely always deprotonated at physiological pH, so a fully protonated head group would not be expected.

(Choices C and D) Phospholipid tails are in the *interior* of the phospholipid bilayer, and therefore are *not* the phospholipid component to which cytosolic molecules bind.

Educational objective:

Phospholipids are composed of a hydrophilic polar head that contains a phosphate group and a hydrophobic tail with one or more fatty acids. Phospholipid properties depend on the head groups attached to the phosphate.

Time spent:0 sec

Subject:
Biochemistry

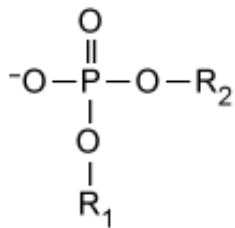
QID:403129

System:
Carbohydrates, Nucleotides, and Lipids

Phospholipids consist of head and tail portions and assemble into bilayer structures through stable interactions of one phospholipid with another. The intracellular and extracellular fluids of eukaryotic cells are separated by a phospholipid bilayer.

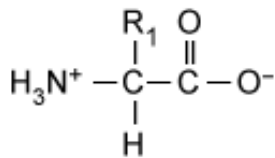
Which structure below is found in phospholipids and prevents simple diffusion of hydrophilic molecules across the cell membrane?

☐ A.



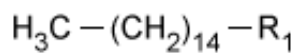
[23%]

☐ B.



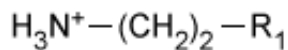
[4%]

☒ C.



[69%]

☐ D.

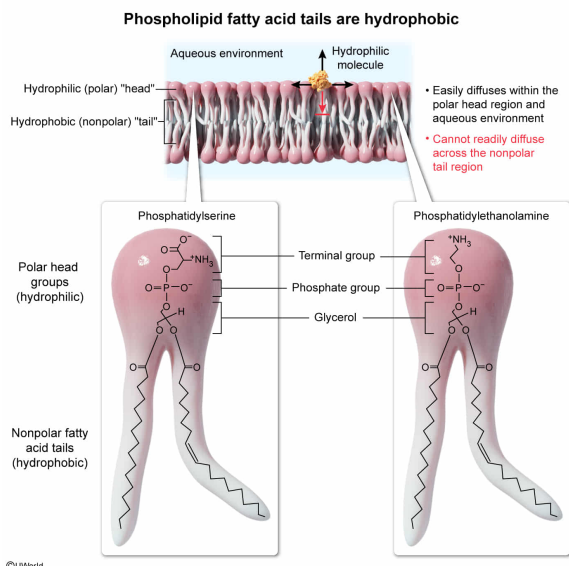


[2%]

Correct

70% answered correctly

Explanation:



Phospholipids are amphipathic molecules (ie, they contain both hydrophilic and hydrophobic regions). The two main regions of phospholipids are:

- A **polar head region**, consisting of **glycerol**, a **phosphate** linked to one of the glycerol carbons, and one of several **additional terminal groups** linked to the phosphate. The terminal head group and the fatty acid tails can vary and give each head group its **unique identity**. **Unequal electron sharing** between atoms in this group yield partial or full charges that interact well with hydrophilic molecules. Hydrophilic molecules easily diffuse within the polar portion of the membrane.
- A **nonpolar tail region** consisting of two **fatty acids** linked to the remaining two carbons of glycerol. The fatty acid tails can be either **saturated or unsaturated** and interact poorly with hydrophilic molecules.

The question asks for the part of a phospholipid that prevents simple diffusion of hydrophilic molecules across the cell membrane. The **nonpolar fatty acid tails** of phospholipids **associate** with each other **in the interior of bilayers**, thereby **forming a hydrophobic region** that prevents hydrophilic molecules from crossing cell membranes in appreciable quantities.

(Choice A) Phosphate groups do not contribute to the hydrophobicity of phospholipids in cell membranes, so they do not prevent hydrophilic molecules from diffusing.

(Choices B and D) The terminal groups shown in these diagrams are hydrophilic and carry charges. As such, they do not contribute to the hydrophobicity of the phospholipid and do not prevent diffusion of hydrophilic molecules.

Educational objective:

Phospholipids consist of a polar head region and a nonpolar tail region. The polar head consists of glycerol, a phosphate, and a terminal group, and is hydrophilic whereas the nonpolar fatty acid tails are hydrophobic. Hydrophilic molecules interact poorly with the hydrophobic tails and cannot easily diffuse across the membrane.

Time spent:0 sec
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 Biochemistry

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System:
 Carbohydrates, Nucleotides, and Lipids